Pokemon Game Agent

By Team 25

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Problem Statement

Motivation

- **Objective:** Develop and train a model that performs beter than a random move selector in Pokemon PvP
- **Inspired by existing papers and models,** including PokeLLMon and Rempton Games' Pokemon Showdown Agents
- Designed as being similar to a battle assistant

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Methodology

Methodology

State Parser
Extracts live battle state from Showdown

Mechanics
Computes damage, KO probabilities, etc.

Heuristics
Simple rule-based evaluation (baseline)

Learning System
Main model for the policy + value network, using MLP

Decisions
Combines mechanics with learning system, recommends moves

Overlay
Frontend UI displaying battle info and control buttons

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Evaluation

Evaluation

```
[eval]      reasons: move order: +75.00; reliability: -4.00;
model bonus: -3.70
[eval] battle_finished tag=battle-gen1randombattle-
2584812700-ulhrnh63ca7gnats2ksbuviqlicxou2pw result=loss
games=100/100
[eval] games=100/100 wins=34 losses=66 ties=0
[eval] complete games=100 wins=34 losses=66 ties=0
win_rate=0.340
[eval] wrote model progress ->
training/artifacts/model_progress.json
Eval result: wins=34 losses=66 ties=0 win_rate=0.340
gate=fail
```

Evaluation log for our first model

- **Evaluation** conducted through battle logs. Keep track of:
  - Battle stats
  - Overall training stats
  - Etc.
- **Metric:** Model’s win-rate

Demo

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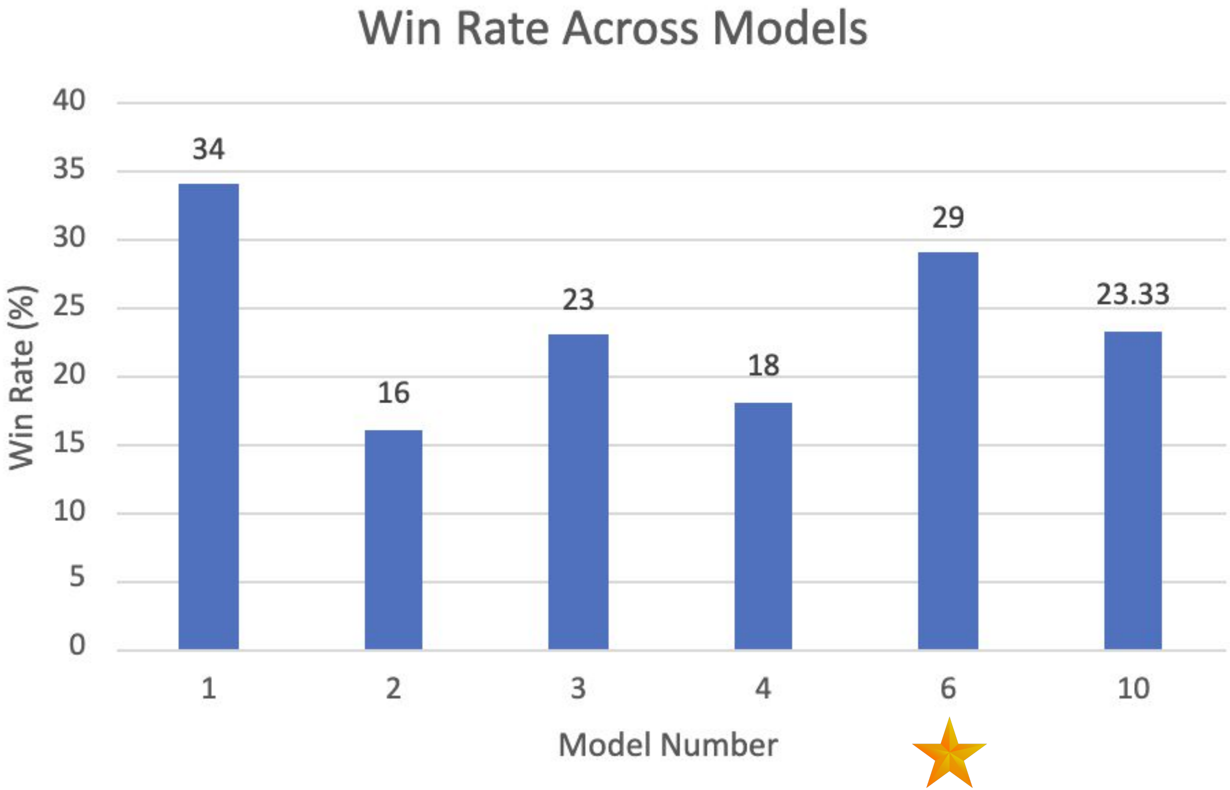
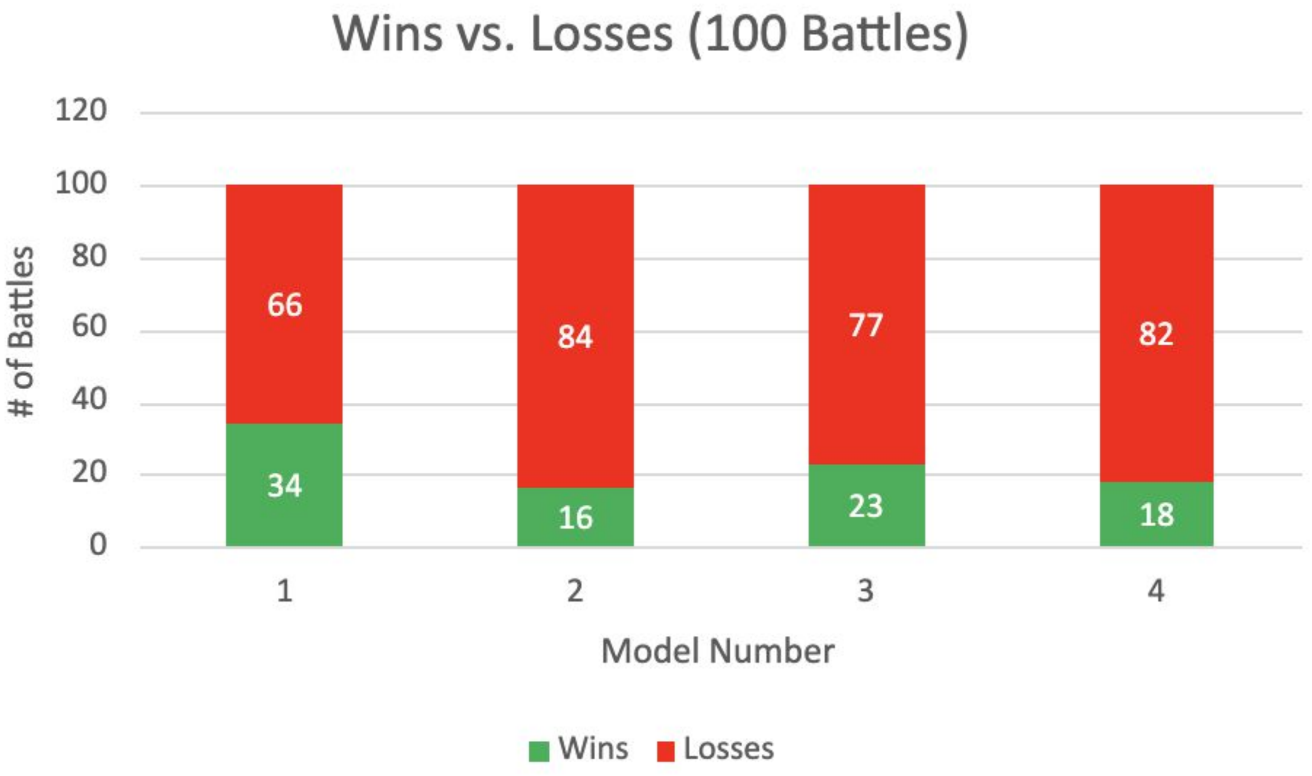
Results & Insights

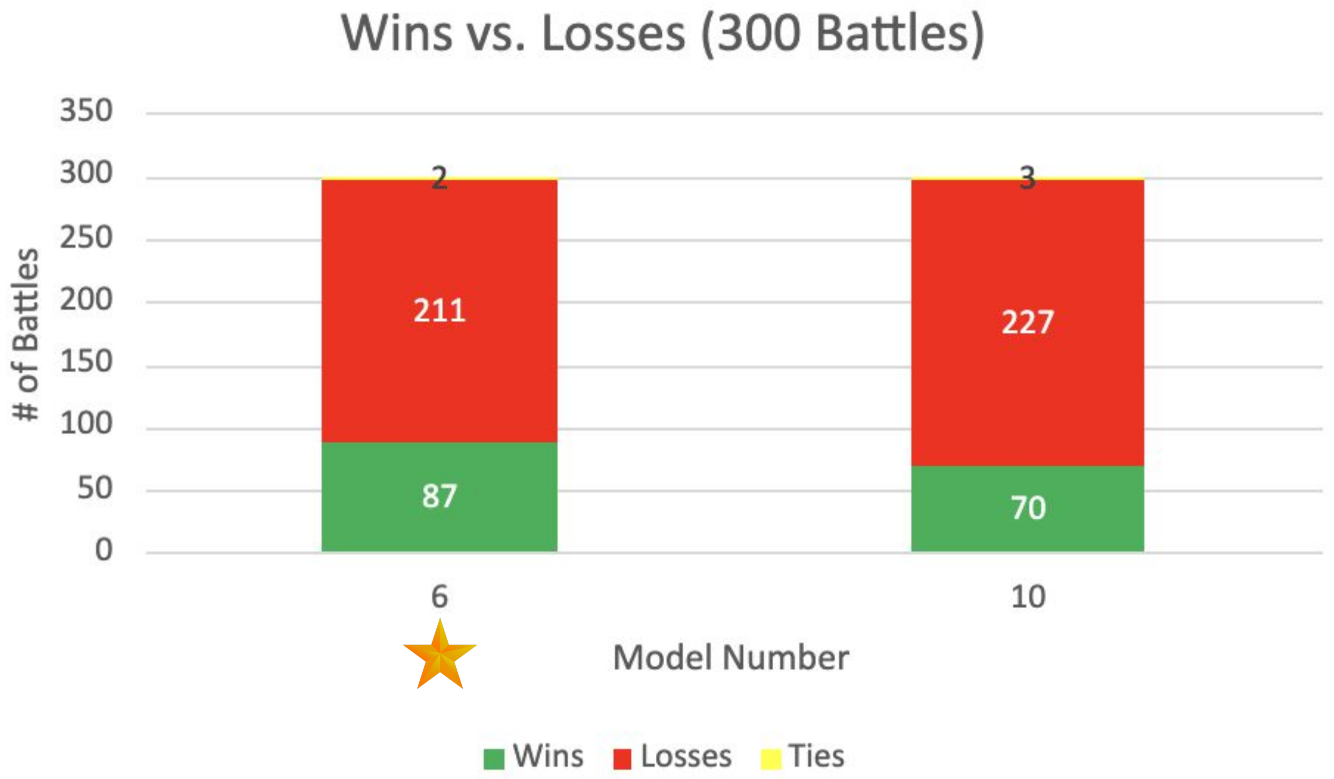
Results and Insights



Results and Insights



Results and Insights



Results and Insights

- Switched from 100 Battles to 300 Battles
  - Search depth was not adjusted (on default value)
  - Opponent response was not implemented properly
  - *Newer models (6 and 10) now utlize opponent prediction based on search*
  - *Results are comparable to existing models!*

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Future Work

Future Work and Improvements

Goal 1

Improve win rate through continued training over the summer

Goal 2

(Possibly) explore other battle formats, such as general random battles

Goal 3

Incorporate more game features, such as held items, into training

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Conclusion

Conclusion

We successfully created an educational recommender system which considers user input and decision making and avoids creating unfair advantages.

We learned the most optimal types of moves based on model reasoning. However, human players often have stronger strategic reasoning due to battle experience.

THANK YOU FOR LISTENING

Does anyone have any questions?